Q1.

The continuous random variable R follows a rectangular distribution with probability density function given by

$$f(r) = \begin{cases} k & -a \le r \le b \\ 0 & \text{otherwise} \end{cases}$$

Prove, using integration, that $E(R) = \frac{1}{2}(b - a)$

(Total 4 marks)

Q2.

Mehreen lives a 2-minute walk away from a tram stop. Trams run every 10 minutes into the city centre, taking 20 minutes to get there. Every morning, Mehreen leaves her house, walks to the tram stop and catches the first tram that arrives. When she arrives at the city centre, she then has a 5-minute walk to her office.

The total time, T minutes, for Mehreen's journey from house to office may be modelled by a rectangular distribution with probability density function

$$f(t) = \begin{cases} 0.1 & a \leqslant t \leqslant b \\ 0 & \text{otherwise} \end{cases}$$

(a) (i) Explain why $a = 27$.

(ii) State the value of b.

(3)

(b) Find the values of E(T) and Var(T).

(2)

(c) Find the probability that the time for Mehreen's journey is within 5 minutes of half an hour.

(2)

(Total 7 marks)

Q3.

Josephine accurately measures the widths of A4 sheets of paper and then rounds the widths to the nearest 0.1 cm. The rounding error, X centimetres, follows a rectangular distribution.

A randomly selected A4 sheet of paper is measured to be 21.1 cm in width.

(a) Write down the limits between which the true width of this A4 sheet of paper lies.

(1)

(b) Write down the value of E(X) and determine the **exact** value of the standard deviation of X.

(3)

(c) Calculate $\mathrm{P}(-0.01 \le X \le 0.03)$.

(1)
(Total 5 marks)

Q4.

(a) The continuous random variable X has a rectangular distribution defined by the probability density function

$$f(x) = \begin{cases} 0.01\pi & u \le x \le 11u \\ 0 & \text{otherwise} \end{cases}$$

where u is a constant.

(i) Show that $u = \frac{10}{\pi}$.

(2)

(ii) Using the formulae for the mean and the variance of a rectangular distribution, find, in terms of π, values for E(X) and Var(X).

(2)

(iii) Calculate **exact** values for the mean and the variance of the circumferences of circles having diameters of length $\left(X + \frac{10}{\pi}\right)$.

(4)

(b) A machine produces circular discs which have an area of Y cm^2. The distribution of Y has mean m and variance 25.

A random sample of 100 such discs is selected. The mean area of the discs in this sample is calculated to be 40.5 cm^2.

Calculate a 95% confidence interval for μ.

(3)
(Total 11 marks)

Q5.

The continuous random variable X has a rectangular distribution defined by

$$f(x) = \begin{cases} k & -3k \le x \le k \\ 0 & \text{otherwise} \end{cases}$$

(a) (i) Sketch the graph of f.

(2)

(ii) Hence show that $k = \frac{1}{2}$.

(2)

(b) Find the **exact** numerical values for the mean and the standard deviation of X.

(3)

(c) (i) Find $\mathrm{P}\left(X \geq -\frac{1}{4}\right)$.

(2)

(ii) Write down the value of $\mathrm{P}\left(X \neq -\frac{1}{4}\right)$.

(1)

(Total 10 marks)

Q6.

(a) The continuous random variable T follows a rectangular distribution with probability density function given by

$$\mathrm{f}(t) = \begin{cases} k & -a \leq t \leq b \\ 0 & \text{otherwise} \end{cases}$$

(i) Express k in terms of a and b.

(1)

(ii) Prove, using integration, that $\mathrm{E}(T) = \frac{1}{2}(b-a)$.

(4)

(b) The error, in minutes, made by a commuter when estimating the journey time by train into London may be modelled by the random variable T with probability density function

$$\mathrm{f}(t) = \begin{cases} \frac{1}{10} & -4 \leq t \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

(i) Write down the value of $\mathrm{E}(T)$.

(1)

(ii) Calculate $\mathrm{P}(T < -3 \text{ or } T > 3)$.

(2)

(Total 8 marks)

Q7.

Students are each asked to measure the distance between two points to the nearest tenth of a metre.

(a) Given that the rounding error, X metres, in these measurements has a rectangular distribution, explain why its probability density function is

$$f(x) = \begin{cases} 10 & -0.05 < x \leq 0.05 \\ 0 & \text{otherwise} \end{cases}$$

(3)

(b) Calculate P(–0.01 < X < 0.02).

(2)

(c) Find the mean and the standard deviation of X.

(2)
(Total 7 marks)

Q8.

The continuous random variable X has a rectangular distribution on the interval (a, $a + k$), where a and k are constants and $k > 0$.

(a) (i) Given that Var(X) = 48, show that $k = 24$.

(2)

(ii) Given also that E(X) = 6, show that $a = -6$.

(2)

(b) Hence calculate P($X > 0$).

(2)
(Total 6 marks)

Q9.

The continuous random variable X has a rectangular distribution on the interval (a, b), where $0 < a < b$.

(a) Given that the mean, μ, is equal to 21 and that the variance, σ^2, is equal to 27, prove that a = 12 and b = 30.

(5)

(b) Hence determine:

(i) P (5 < X < 20);

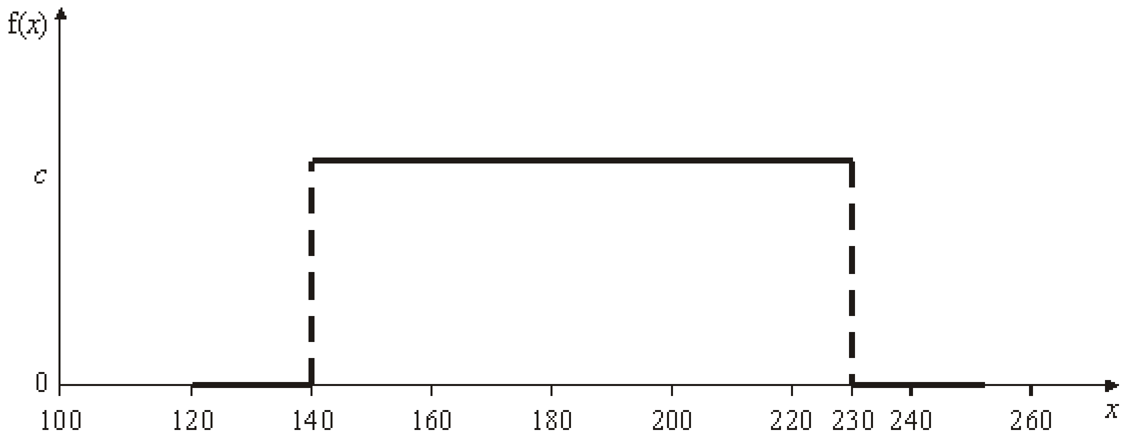
(3)

(ii) $\mathrm{P}\left(X < \mu - \dfrac{\sigma\sqrt{3}}{2}\right)$.

(3)
(Total 11 marks)

Q10.

The probability density function, f(x), for the weight, X grams, of a banana is shown by the following graph.



(a) Find the value of the constant c, as used on the vertical axis.

(1)

(b) Determine $P(X < 200)$.

(2)

(c) Find values for the mean and variance of X.

(2)
(Total 5 marks)

Q11.

The continuous random variable X has a rectangular distribution over the interval c to $7c$, where c is a positive constant.

(a) (i) Find, in terms of c, expressions for the mean, μ, and variance, σ^2, of X.

(2)

(ii) Hence show that $E(X^2) = 19c^2$.

(2)

(b) Given that $E(X^2) = 171$, determine the value of c.

(1)

(c) Determine:

(i) $P\left(X > \frac{\mu}{2} + \frac{\sigma}{\sqrt{3}}\right)$;

(3)

(ii) the value of d such that $P(X < d) = 0.25$.

(3)
(Total 11 marks)

Q12.

The error, X grams, made by a butcher's scales, may be modelled by a continuous uniform (rectangular) distribution defined by:

$$f(x) = \begin{cases} \frac{1}{4c} & -c \le x \le 3c \\ 0 & \text{otherwise.} \end{cases}$$

(a) State, in terms of c, values for the mean and variance of X.

(2)

(b) Given that $E(X^2) = \frac{28}{3}$, show that $c = 2$.

(3)

(c) Determine:

(i) $P(X = 2)$;

(1)

(ii) $P(|X| < 2.5)$.

(3)

(Total 9 marks)

Q13.

A video recorder clock is corrected automatically by a signal every 24 hours. The deviation, T seconds, of the clock from the correct time, immediately prior to correction, may be modelled by the following probability density function.

$$f(t) = \begin{cases} \frac{1}{5b} & -b < t < 4b \\ 0 & \text{otherwise} \end{cases}$$

(a) Given that $E(T) = 9$, show that $b = 6$.

(2)

(b) Hence calculate $P(T < 15)$.

(2)

(Total 4 marks)

Q14.

The continuous random variable X has a rectangular distribution over the interval a to $a + k$, where a and k are positive constants.

(a) Find, in terms of a and k, expressions for the mean and variance of X .

(2)

(b) Given that $E(X) = 21$ and $\text{Var}(X) = 3$, show that $k = 6$ and $a = 18$.

(3)

(c) Hence calculate $P(20 < X < 25)$

(3)

(Total 8 marks)

Q15.

The lengths, X centimetres, of boards used in garden fencing are uniformly distributed over the interval 198 to 204.

(a) Find values for the mean, μ, and the variance, σ^2, of X.

(2)

(b) Write down the probability density function, f(x), of X for $198 \le x \le 204$.

(1)

(c) Determine:

(i) $P(X = \mu)$;

(1)

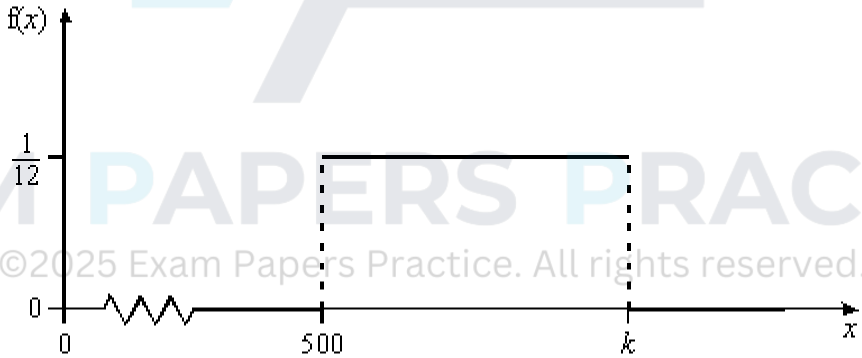
(ii) $P(X < 200 - \sigma)$.

(3)

(Total 7 marks)

Q16.

The probability density function, f(x), for the weight, X grams, of marshmallow pieces in a tub is shown by the following graph.



(a) State why the value of k on the horizontal axis is 512.

(1)

(b) Determine values for the mean, μ, and standard deviation, u, of X.

(3)

(c) Calculate:

(i) $P(X < 509)$;

(2)

(ii) $P(X > \mu - \sigma)$.

(3)

(Total 9 marks)

Q17.

A supermarket sells carrots in bags marked "2 kilograms". The amount, in grams, by which the weight of carrots in a bag differs from 2 kilograms may be modelled by the random variable, X, with probability density function

$$f(x) = \begin{cases} \dfrac{1}{3c} & -c < x < 2c \quad \text{where } c \text{ is a constant} \\ 0 & \text{otherwise} \end{cases}$$

(a) Write down the mean of X in terms of c.

(1)

(b) Using integration, show that $E(X^2) = c^2$.

(4)

(c) Find the standard deviation of X in terms of c.

(3)

(d) The carrots in a large sample of bags are weighed. For these bags, the mean value of X is found to be 22. Estimate:

(i) the standard deviation of X ;

(4)

(ii) the minimum value of the weight of carrots in a bag.

(2)

(Total 14 marks)

Q18.

The errors, in grams, made by a butcher's scales may be modelled by a random variable, y, with probability density function

$$f(y) = \begin{cases} k & -6 < y < 14 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that $k = 0.05$.

(2)

(b) Write down the mean value of Y.

(1)

(c) (i) Use integration to find $E(Y^2)$.

(3)

(ii) Hence, or otherwise, find the standard deviation of Y.

(3)

(d) (i) Find the probability that an error is negative.

(1)

(ii) Show that the probability of an error having magnitude less than 2 is 0.2.

(2)

(iii) Find the probability that the magnitude of an error is greater than 4.

(2)
(Total 14 marks)

Q19.

A supermarket sells 1 kg bags of potatoes. It is not possible to select potatoes so that each bag weighs exactly 1 kg and so the weight of each bag exceeds 1 kg by an amount, x grams. It is believed that X may be modelled by a random variable which follows a rectangular distribution with probability density function:

$$f(x) = \begin{cases} \frac{1}{c} & 0 \le x \le c \\ 0 & \text{otherwise} \end{cases}$$

where c is a constant.

(a) (i) Write down the mean of this distribution in terms of c.

(ii) Using integration, show that the variance is $\frac{c^2}{12}$.

(5)

(b) Find the probability that X is greater than $\frac{c}{4}$.

(2)

(c) The mean weight of a random sample of bags exceeds 1 kg by 14 grams.

Estimate:

(i) the standard deviation of X;

(4)

(ii) the weight of the heaviest bag.

(1)
(Total 12 marks)